MATH INQUIRERS HOLIDAY PROGRAM

Question Bank

Three tiers of questions for every lesson, with answers

Notice it. Test it. Explain why.

Question Bank • Ages 8–10 (builds on Primary 3 maths)

Math Inquirers P3 Ages 8-10. Notice it. Test it. Explain why.

How to use this bank

- Each lesson has three tiers: Start here, Think harder and Go further. They are open to everyone. Nobody is placed in a tier: every child can start with Start here, and any child may try Think harder or Go further at any time. Go further questions are good ones to talk about together.
- START HERE questions are approachable for a child who has not yet been taught the topic. They can start with objects, a drawing, counting, guess and check or skills from earlier years, and they use no unexplained words. Use them for warm-ups and to build confidence.
- THINK HARDER questions need the idea taught in the lesson, used in a new setting: two-step problems, patterns with a hidden rule, unknowns and larger numbers.
- GO FURTHER questions need the topic well understood, plus an extension: find every way, work backwards, test a case, justify an answer, change the problem or make up your own. This program is inspired by research-based approaches used in gifted education; it is not an official GEP standard and is not affiliated with MOE.
- Ask for the reason every time: "How do you know?" A correct answer with no reason is only half done.
- Answers are at the back. Do not hand the answer pages to children.

What each lesson practises

Lesson 1	Number Sleuths (Numbers to 10 000)	6 + 8 + 9 questions
Lesson 2	Cryptarithm Crackers (Addition and Subtraction)	6 + 9 + 9 questions
Lesson 3	Times Table Treasure (Tables of 6, 7, 8 and 9)	6 + 8 + 9 questions
Lesson 4	Model Makers (Word Problems: Bar Models)	6 + 9 + 9 questions
Lesson 5	Remainder Riddles (Division)	6 + 8 + 9 questions
Lesson 6	Fraction Walls (Fractions)	6 + 9 + 9 questions
Lesson 7	Area Architects (Area and Perimeter)	6 + 8 + 9 questions
Lesson 8	Angle and Line Detectives (Angles, Perpendicular and Parallel Lines)	6 + 8 + 9 questions
Lesson 9	Measure and Money Masters (Length, mass, volume, money and time)	6 + 9 + 9 questions
Lesson 10	Puzzle Day (Mixed puzzles: bar graphs and logic)	6 + 8 + 9 questions

Lesson 1: Number Sleuths

Numbers to 10 000 • Week 1

START HERE (open with objects, drawing or counting)

S1. Write the number. Think of 6 big boxes of 1000 sweets, 4 packets of 10 sweets and 9 loose sweets. (There are no boxes of 100.)

S2. Look at the place value chart: Th 2 | H 8 | T 3 | O 7. This is the number 2837. How much is the digit 8 worth?

S3. Use the cards 9, 1, 4, 6 once each. Make the smallest 4-digit number.

S4. What number is 10 less than 5003?

S5. Continue the pattern: 7250, 7500, 7750, ___, ___

S6. Order from smallest to greatest: 4089, 4809, 4098, 4890

THINK HARDER (the lesson idea in a new setting)

T1. Use the cards 5, 0, 8, 2 once each. Make the greatest even number and the smallest odd number.

T2. How many 4-digit numbers can you make with the cards 1, 1, 0, 0? List them.

T3. A number has 7 thousands, 13 hundreds and 6 ones. What is the number?

T4. I am between 5000 and 6000. My hundreds, tens and ones digits are all the same. My digits add up to 11. What number am I?

T5. What is the smallest 4-digit number with all different digits that add up to 10?

T6. How many times do you write the digit 5 when you write all the numbers from 1 to 100?

T7. Continue the pattern: 1, 3, 7, 15, 31, ___, ___

T8. How many 4-digit numbers between 3000 and 4000 read the same forwards and backwards, like 3443?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ali writes a 4-digit number using only odd digits, all different. It is as close to 5000 as possible. What is it?

G2. A pattern starts at 1000 and goes up by 7 each time: 1000, 1007, 1014, ... Is 1100 in the pattern? Explain.

G3. How many digits are needed to number the pages of a book from page 1 to page 120?

G4. 4-digit numbers whose digits add up to 3 can be counted by their first digit: 1, 3, 6. Count the ones whose digits add up to 4 by their first digit too. Use the pattern to predict how many 4-digit numbers have digits that add up to 5, and check by listing the ones that start with 4.

G5. I am an even 4-digit number with all different digits. My thousands digit is 4 more than my ones digit. My hundreds digit is double my tens digit. My digits add up to 20. What number am I?

G6. Mei makes 3-digit numbers using any three of the cards 2, 4, 6, 8. How many different numbers can she make?

G7. Make up your own "What number am I?" puzzle. Choose a secret 4-digit number and write 3 or 4 clues about its digits. Check that ONLY ONE number fits all your clues. Then swap with a partner.

G8. A 4-digit number has all different digits. Its thousands digit is the sum of its other three digits. What is the greatest such number? What is the smallest?

G9. Mei uses the cards 1, 2, 3, 4 once each to make 4-digit numbers. She lists them all from smallest to greatest. What is the 10th number in her list?

Lesson 2: Cryptarithm Crackers

Addition and Subtraction • Week 1

START HERE (open with objects, drawing or counting)

S1. A library has 3999 storybooks and 2456 comic books. How many books are there altogether? Write the numbers in a Th H T O grid and add the ones first.

S2. A shop has 7000 stickers and sells 2468 of them. How many are left? Hint: draw a number line from 2468 to 7000 and jump to 2500, then to 3000, then to 7000. Add up the jumps.

S3. Find the missing digit. Add in columns, starting with the ones, or try the digits one at a time. 4□27 + 1365 = 5992

S4. Find the missing digit. Try 5, 6 and 7 and check each one by taking away. 8□50 − 3420 = 5230

S5. A magic square has 3 rows. Every row adds up to the same total, 2400. One row shows 900, 600 and a gap. What number goes in the gap?

S6. A shop had 5000 stickers. It sold 1275 on Monday and 1480 on Tuesday. Draw a bar for the 5000 stickers and cut it into the two sold parts and the part left. How many stickers are left?

THINK HARDER (the lesson idea in a new setting)

T1. Find the missing digits (each letter is one digit; turn it round and add): 7A03 − B46C = 2538

T2. Find the missing digits (each letter is one digit): 2A7B + C6D9 = 6000

T3. In a number pyramid, each brick is the sum of the two bricks under it. The bottom row is 1200 and two empty bricks. The middle row is 2650 and one empty brick. The top brick is 6000. Find the three empty bricks.

T4. Ali meant to add 1099 to a number, but he added 1999 by mistake and got 6543. What is the correct answer?

T5. A zoo had 6000 visitors in three days. On Monday there were 1874 visitors. On Tuesday there were 650 more than on Monday. How many visitors came on Wednesday? Check your answer by adding all three days.

T6. AB + B = BA. Different letters are different digits. Find A and B.

T7. Siti worked out 9999 − 3746 = 6253 by taking each digit away from 9. Use her answer to work out 10 000 − 3746 and 10 003 − 3746 without a new column subtraction. How do you know?

T8. AA + BB = CBC. Different letters are different digits. Find them.

T9. Use the cards 2, 3, 5, 6 once each to make two 2-digit numbers. Make the difference as small as possible.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Find the missing digits: 8A0B − C4D7 = 2468

G2. A 4-digit number has all different digits that add up to 30. Find the difference between the greatest and the smallest such number.

G3. AA + BB + CC = ABC. Different letters are different digits. Find them.

G4. ABC − CB = CAB. Different letters are different digits. Find the digits.

G5. Make a magic square with the numbers 1000, 1100, 1200, 1300, 1400, 1500, 1600, 1700 and 1800. What is the magic total?

G6. Use the digits 1 to 9 once each to make a sum of two 3-digit numbers with a 3-digit answer. What is the greatest possible answer?

G7. Make up your own missing-digits sum. Write a 4-digit + 4-digit addition that works, then rub out exactly 3 digits in 3 DIFFERENT columns and write the letters A, B and C instead. Swap with a partner. Does your partner find only one answer? Explain why your sum has only one answer.

G8. Use the digit cards 1 to 8 once each to make two 4-digit numbers. Make their difference as small as possible.

G9. ABC + ACB = CBA. Different letters are different digits. Find them.

Lesson 3: Times Table Treasure

Tables of 6, 7, 8 and 9 • Week 2

START HERE (open with objects, drawing or counting)

S1. Draw 6 rows of 8 dots. How many dots are there? Now turn your drawing sideways: it shows 8 rows of 6 dots. How many dots now? What do you notice?

S2. You know 5 × 9 = 45. Use it to find 6 × 9.

S3. 63 apples are packed into boxes of 7 apples with none left over. Count in 7s (7, 14, 21, ...) to find how many boxes there are.

S4. A spider has 8 legs. How many legs do 7 spiders have?

S5. Count in 6s from 6 up to 60 (6, 12, 18, ...). Write only the LAST digit of each number. Which last digits do you see?

S6. Find 9 × 7 by working out 10 × 7 and taking away one 7.

THINK HARDER (the lesson idea in a new setting)

T1. The digits of every answer in the 9 times table (9 × 1 to 9 × 10) add up to 9. Which of these are in the 9 times table: 54, 65, 72, 83?

T2. What is the last digit of 8 × 8 × 8? Use last digits only.

T3. Ravi says: "Every answer in the 6 times table is even." Is he right? Explain why.

T4. Find ALL the 2-digit numbers that can be divided exactly by 7 and whose digits add up to 10.

T5. Compare each pair. Which is more, and by how much? (a) 7 × 7 and 6 × 8 (b) 8 × 8 and 7 × 9 (c) 7 × 8 and 6 × 9. What do you notice?

T6. The last digit of 9 × 9 is 1. What is the last digit of 9 × 9 × 9 × 9?

T7. Ben multiplies a number by 6. His answer ends in 4. What could the last digit of his number be? Find all the possible digits.

T8. Find three different ways to work out 8 × 9 using facts you already know (for example 5 × 9 plus something). Write each way and the answer.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Four numbers in a multiplication square are 42 and 48 in the top row, and 49 and 56 in the row below. What are the two row numbers and the two column numbers?

G2. Ben writes the answers of the 6 times table from 6 × 1 to 6 × 20. How many of the answers end in 8?

G3. How many numbers from 1 to 200 can be divided exactly by both 6 and 9?

G4. In the 8 times table, which digits can NEVER be the last digit of an answer? Explain why. Is the same true for the 6 times table?

G5. Siti says the answer to 6 × 7 × 8 × 9 ends in 4. Is she right? Use last digits only.

G6. Mei has fewer than 100 sweets. She can share them equally among 6 friends, among 9 friends and among 4 friends, with none left over each time. Find ALL the numbers of sweets she could have. What do you notice about the answers? Why?

G7. Make up your own riddle: "I am a 2-digit number in the ___ times table ..." Write 2 or 3 clues (about odd or even, the digits, being between two numbers). Check that only ONE number fits, by listing the table. Swap with a partner.

G8. Ben keeps multiplying by 7: 7, 49, 343 … He writes only the last digit each time: 7, 9, 3 … What is the last digit when he has multiplied 8 sevens together? Explain the pattern.

G9. I am a 3-digit number less than 200. I can be divided exactly by 7 and by 9. My digits add up to 9. What number am I?

Lesson 4: Model Makers

Word Problems: Bar Models • Week 2

START HERE (open with objects, drawing or counting)

S1. There are 2345 people at a concert. 1180 of them are adults. How many are children? Draw one long bar for all the people and mark off the part for the adults.

S2. A red ribbon is 145 cm long. A blue ribbon is 68 cm longer. Draw the red ribbon as a bar, then draw a longer bar for the blue ribbon. How long is the blue ribbon?

S3. Ravi has 512 stamps. Siti has 139 fewer stamps than Ravi. Draw a bar for each child. How many stamps does Siti have? How many do they have altogether?

S4. Ali has 23 cards. Ben has 4 times as many cards as Ali. Draw 1 box for Ali and 4 boxes for Ben. How many cards does Ben have? How many do they have altogether?

S5. A box has 1000 beads. Mei uses 365 beads for a necklace and 248 beads for a bracelet. How many beads are left?

S6. A pen costs $3. A bag costs 5 times as much as the pen. Draw 1 box for the pen and 5 boxes for the bag. How much more does the bag cost than the pen?

THINK HARDER (the lesson idea in a new setting)

T1. Ali and Ben have 96 cards altogether. Ali has 3 times as many cards as Ben. How many more cards does Ali have than Ben?

T2. Siti has 54 more beads than Mei. Siti gives Mei some beads, and now they have the same number. How many beads did Siti give Mei?

T3. Three boxes hold 600 pens altogether. Box A has 50 more pens than Box B. Box C has 50 more pens than Box A. How many pens are in Box C?

T4. A bag of rice and a bag of flour weigh 9 kg altogether. The rice is 3 kg heavier than the flour. How heavy is each bag?

T5. Mei is 8 years old. Her mother is 36. How old will Mei be when her mother is 3 times as old as Mei?

T6. Ali has $45. Ben has $18. Grandma gives each boy the same amount of money. Now Ali has twice as much money as Ben. How much money did Grandma give each boy?

T7. Ben had 3 times as many marbles as Ravi. Ben gave Ravi 20 marbles, and then they had the same number. How many marbles did Ben have at first?

T8. A bag of sugar and 2 bags of flour have a total mass of 1700 g. Each bag of flour is 400 g heavier than the bag of sugar. What is the mass of the bag of sugar?

T9. Ali has saved $48. Ben has saved $15. From now on each boy saves $6 a week. After how many weeks will Ali have twice as much as Ben?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ali and Ben have 120 stickers altogether. Ali has 3 times as many as Ben. Ali gives some stickers to Ben, and then Ali has only twice as many as Ben. How many stickers does Ali give to Ben?

G2. A pencil box and 3 pens cost $25 altogether. A pencil box and 5 pens cost $35 altogether. How much does one pen cost? How much does the pencil box cost?

G3. Ali, Ben and Ravi have 185 marbles altogether. Ali has twice as many as Ben. Ben has 15 more than Ravi. How many marbles does each boy have?

G4. Ravi, Mei and Siti have 160 beads altogether. Mei has 3 times as many beads as Siti. Ravi has 20 more beads than Mei. How many beads does each child have?

G5. A club has 3 times as many boys as girls. Then 12 boys leave and 12 girls join, and now there are the same number of boys and girls. How many children are in the club?

G6. 2 jugs and 3 cups hold 1800 ml of water altogether. A jug holds 3 times as much as a cup. How much does a cup hold?

G7. Make up your own before-and-after story about stickers or money, where one child gives some to another child. Draw its model, solve it and check it. Then change ONE number in your story. What changes in the answer, and what stays the same? Swap with a partner.

G8. Ravi is 7 years old and his sister is 3. In how many years will their ages add up to 30? Explain how you know.

G9. A ribbon 4 m 75 cm long is cut into 3 pieces. The second piece is twice as long as the first. The third piece is 25 cm longer than the second. How long is the third piece?

Lesson 5: Remainder Riddles

Division • Week 3

START HERE (open with objects, drawing or counting)

S1. Share 35 counters equally among 4 children. How many counters does each child get? How many counters are left over? (Use counters or draw dots in 4 groups.)

S2. Put 58 sweets into bags with 7 sweets in each bag. How many full bags can you fill? How many sweets are left over? (Use counters or draw dots.)

S3. 6 friends share 45 stickers equally. How many does each friend get? How many are left over?

S4. 30 children go on an outing. Each car holds 4 children. How many cars are needed?

S5. A ribbon is 50 cm long. Siti cuts it into pieces 6 cm long. How many whole pieces does she get? How much ribbon is left?

S6. Ben packs muffins into boxes of 5. He fills 7 boxes and has 3 muffins left over. How many muffins does he have?

THINK HARDER (the lesson idea in a new setting)

T1. Mei has 53 beads. She makes bracelets with 8 beads each. How many bracelets can she make? How many more beads does she need to make one more bracelet?

T2. I am a number between 20 and 30. When I am divided by 4, the remainder is 3. When I am divided by 6, the remainder is 5. What number am I?

T3. What is the biggest 2-digit number that leaves a remainder of 8 when it is divided by 9?

T4. A boat can carry 9 people at a time. 100 people want to cross the river. How many boatloads are needed?

T5. Siti fills 9 bags with 7 oranges each and has 5 oranges left over. She packs all the oranges again into bags of 8. How many full bags does she get? How many are left over?

T6. A pattern goes circle, triangle, square, circle, triangle, square, … What is the 50th shape?

T7. Find all the 2-digit numbers less than 40 that leave a remainder of 2 when divided by 3 AND when divided by 4.

T8. Mei has 29 beads. She puts them into bags of 4. Then she tips them out and puts them into bags of 5. Then into bags of 6. Each time she writes how many beads are left over. Which bag size leaves the most beads over? Which leaves the fewest?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. I am a 2-digit number. When I am divided by 3, by 4 or by 5, the remainder is always 2. What number am I?

G2. A box has more than 30 but fewer than 100 chocolates. They can be shared equally among 2, 3, 4 or 6 children with none left over. When they are shared among 5 children, 2 are left over. How many chocolates are there?

G3. A number is divided by 6. The answer and the remainder are the same number, and it is not 0. Find all the numbers it could be.

G4. Ben counts his marbles in 5s and has 3 left over. He counts them in 7s and has 3 left over. He has more than 50 and fewer than 100 marbles. How many does he have?

G5. When 100 is divided by a 1-digit number, the remainder is 1. Find all the numbers it could be.

G6. A farmer counts her eggs. In 2s, 3s, 4s, 5s or 6s there is always 1 egg left over. In 7s there are none left over. What is the smallest number of eggs she can have?

G7. Make up your own remainder riddle: "I am a number between ___ and ___. When I am divided by ___ the remainder is ___. When I am divided by ___ the remainder is ___." Check that only ONE number fits, by listing the numbers for the first clue and testing them. Swap with a partner.

G8. Today is Monday. What day of the week will it be 100 days from today? What day of the week was it 100 days ago? Explain how you know.

G9. Mr Lim gives pens to his pupils. If each pupil gets 4 pens, 10 pens are left over. If each pupil gets 6 pens, he is 4 pens short. How many pupils are there? How many pens does he have?

Lesson 6: Fraction Walls

Fractions • Week 3

START HERE (open with objects, drawing or counting)

S1. A chocolate bar has 10 equal pieces. Ali eats half of the bar. How many pieces is that? Write it as 1/2 = ?/10. (Draw the bar and cut it in two halves if you like.)

S2. Two pizzas are the same size. One is cut into 6 equal slices and the other into 9 equal slices. Which pizza has the bigger slices? So which is bigger, 1/6 of a pizza or 1/9 of a pizza?

S3. A bar is cut into 9 equal parts. Shade 2 parts. Then shade 5 more parts. What fraction of the bar is shaded now?

S4. A bar is cut into 8 equal parts and 3 parts are shaded. What fraction of the bar is NOT shaded? (Draw it and count.)

S5. Put these in order, smallest first: 5/12, 1/12, 11/12.

S6. A bar is cut into 5 equal parts and 2 parts are shaded (2/5). Now cut every part into 2 equal pieces. How many small pieces are shaded now, out of how many pieces? Write it as 2/5 = ?/10.

THINK HARDER (the lesson idea in a new setting)

T1. Find ALL the fractions with bottom number 12 that are bigger than 1/2 but smaller than 3/4. Hint: look at the twelfths row of your wall. Which one is equal to 2/3?

T2. Which is bigger, 3/5 or 7/10? Show how you know.

T3. Siti reads 1/4 of a book on Monday and 1/2 of the book on Tuesday. What fraction of the book has she read? What fraction is left?

T4. Put these in order, smallest first: 2/3, 5/6, 7/12.

T5. Write three different fractions that are all equal to 2/3, each with a different bottom number. How do you know they are equal?

T6. A jug is 5/6 full of water. Ravi pours out 1/2 of the whole jug. What fraction of the jug is still full?

T7. Which fraction with bottom number 8 is exactly halfway between 1/4 and 1/2?

T8. Work out 1/2 + 1/4 + 1/8.

T9. Mei says 2/3 is bigger than 3/4, because 2/3 is missing only one piece. Is she right? Explain.

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Which is bigger, 3/7 or 5/9? Use 1/2 to help you.

G2. Which is bigger, 2/5 or 4/9? Hint: make the top numbers the same.

G3. Use the digits 1, 2, 3 and 6, once each, to make two equal fractions that are each less than 1: ?/? = ?/?. Find both ways.

G4. Two fractions both have 12 as the bottom number. Together they make 5/6. One of them is 1/6 more than the other. What are the two fractions?

G5. A fraction is equal to 2/3. Its bottom number is 4 more than its top number. What is the fraction?

G6. I am a fraction equal to 3/4. My top number and my bottom number add up to 14. What fraction am I?

G7. Make up your own "What fraction am I?" riddle, like: "I am equal to 3/4. My top and bottom numbers add up to ___." Choose the number so that exactly ONE fraction fits. Check it, then swap with a partner. Is your riddle always going to work for any number you choose? Explain.

G8. Fill one whole bar using exactly 5 pieces. You may use 1/2, 1/4 and 1/8 pieces, as many of each as you like. Find all the ways.

G9. Find a fraction that is bigger than 2/3 but smaller than 3/4. Hint: the fraction wall has no row that helps, so cut every part into smaller equal parts. Can you find another one?

Lesson 7: Area Architects

Area and Perimeter • Week 4

START HERE (open with objects, drawing or counting)

S1. A rectangle is 9 cm long and 4 cm wide. Walk all the way round its edge: how far is that? (This distance is called the perimeter.) Draw it on squared paper if you like.

S2. Cover a rectangle 7 cm long and 6 cm wide with squares that are 1 cm by 1 cm. There are 6 rows with 7 squares in each row. How many squares cover it? (The number of squares is the area, in cm^2.)

S3. A square has 4 equal sides. The distance all the way round it is 32 cm. How long is each side?

S4. A square floor is covered with 36 small squares (1 cm by 1 cm), with the same number of squares in each row and each column. How many squares are there along one side? How far is it all the way round the edge?

S5. A rectangle is covered by 40 small squares (1 cm by 1 cm), with 8 squares in each row. How many rows are there? How far is it all the way round the edge?

S6. A figure is made of 7 squares, each 1 cm by 1 cm. What is its area? Can you say its perimeter without seeing it?

THINK HARDER (the lesson idea in a new setting)

T1. Find all the rectangles with whole-cm sides and an area of 24 cm^2. Which one has the shortest perimeter?

T2. A rectangle is 12 cm long. Its perimeter is 30 cm. Find its area.

T3. A rectangle 8 cm by 2 cm and a square have the same perimeter. Which has the bigger area, and by how much?

T4. A square with sides of 6 cm is cut into two rectangles of the same size. What is the perimeter of each rectangle?

T5. How many tiles 2 cm by 2 cm are needed to cover a mat 10 cm by 8 cm?

T6. A rectangle has an area of 18 cm^2 and a perimeter of 18 cm. Find its length and width.

T7. Mei's garden is 9 m by 5 m. Ali's garden is 7 m by 7 m. Whose garden needs the longer fence? Whose has the bigger area?

T8. Use exactly 6 squares (each 1 cm by 1 cm), whole sides touching. Draw a figure with a perimeter of 10 cm and one with a perimeter of 14 cm. What is the biggest perimeter possible?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. List every rectangle with whole-cm sides and a perimeter of 24 cm, with its area, in a table. What do you notice as the rectangle gets closer to a square? Use what you notice to predict the biggest area for a perimeter of 40 cm, then check.

G2. A rectangle is made of 2 same-size squares side by side. Its perimeter is 36 cm. What is its area?

G3. A rectangle 6 cm by 4 cm is cut into two rectangles with one straight cut that is parallel to one side. Find the total perimeter of the two pieces. Is there more than one answer?

G4. A square has a perimeter of 20 cm. Four of these squares are put together to make one big square. Find the perimeter and the area of the big square. The perimeter of the big square is not 4 times as long as 20 cm. Explain why, and why the area is 4 times as big.

G5. A rectangle is 10 m long and 8 m wide. A 4 m by 3 m rectangle is cut out of one corner. Find the area and the perimeter that are left. Then cut the same 4 m by 3 m rectangle out of the MIDDLE of one long side instead. Find the new area and perimeter. What stays the same and what changes?

G6. A square with sides of 8 cm is cut into four same-size squares. The four small squares are put in a row, touching whole sides. Find the perimeter of the row. Then compare with the perimeter of the big square and with the area of each shape. What stays the same and what changes?

G7. Make up your own rectangle puzzle. Choose a rectangle with whole-cm sides, then give a partner only TWO clues about it (for example its area and its perimeter). Check that only ONE rectangle fits both clues. Which two clues did you choose, and why do they work?

G8. Ali joins 5 same-size squares with sides of 2 cm, whole sides touching. What is the biggest perimeter his shape can have? The smallest?

G9. A rectangle 7 cm by 5 cm has a 1 cm square cut out of each of its 4 corners. Find the new area and the new perimeter.

Lesson 8: Angle and Line Detectives

Angles, Perpendicular and Parallel Lines • Week 4

START HERE (open with objects, drawing or counting)

S1. A right angle is a square corner, like the corner of a page. How many right angles does a rectangle have? Check with the corner of a piece of paper.

S2. At 3 o'clock the minute hand points at 12 and the hour hand points at 3. Is the angle between the hands smaller than, bigger than, or the same as the corner of a page?

S3. At 2 o'clock, is the angle between the clock hands smaller or bigger than the corner of a page?

S4. Parallel lines are straight lines that stay the same distance apart and never meet, like the two rails of a train track. Name 2 capital letters that have a pair of parallel lines.

S5. Two straight lines cross like a plus sign (+). Each square corner where they cross is a right angle. How many right angles do they make?

S6. 3 lines go out from one point, spread out like a fan. An angle is the space between two lines. How many different angles can you find? (Draw it and count.)

THINK HARDER (the lesson idea in a new setting)

T1. I have 4 sides and 4 right angles. My sides are not all the same length. What shape am I? How many pairs of parallel lines do I have?

T2. A U shape is made of 5 squares: 3 squares in a row with 1 more square on top of each end square. How many right angles are inside it?

T3. Look at a clock face. For each o'clock time from 1 o'clock to 11 o'clock, is the angle between the hands (the short way round) smaller than, the same as, or bigger than a right angle? Put each time in the right group.

T4. The # sign is made of 4 lines: 2 upright lines and 2 lines across. How many pairs of parallel lines does it have? How many pairs of perpendicular lines?

T5. Draw me on square grid paper: (a) I have 4 sides, exactly 2 right angles and exactly 1 pair of parallel lines. (b) I have 4 sides and 2 pairs of parallel lines, but no right angles.

T6. How many right angles can you find in the letter E (4 straight lines)?

T7. Is the smaller angle between the clock hands at 4 o'clock the same size as at 8 o'clock?

T8. The letter H is made of 3 straight lines: 2 upright lines and 1 line across. List every pair of lines. Which pairs are parallel and which pairs are perpendicular?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. 4 lines from one point make 6 angles. How many lines from one point are needed to make exactly 10 angles?

G2. A staircase is made of columns of 1, 2 and 3 squares standing side by side (3 steps). It has 6 right angles inside it. How many right angles are inside a staircase with 4 steps? With 5 steps? With 10 steps? Find a rule and explain why it works.

G3. A square has 2 lines down and 2 lines across inside it, like a noughts-and-crosses grid. How many right angles can you find?

G4. A 4-sided shape has 3 right angles. Draw it. Test the 4th angle with your checker. Can a 5-sided shape have exactly 3 right angles? Draw one.

G5. Mei says: "Perpendicular lines must be one going straight up and one going straight across." Is she right? On a square grid, draw two perpendicular lines that are both slanted. Check with your checker. How do you know they are perpendicular?

G6. Ben says: “A shape with 4 sides and 2 pairs of parallel lines must have right angles.” Is he right?

G7. Make up your own "Which shape am I?" riddle about a shape on square grid paper. Use words like sides, right angles, parallel and perpendicular, and give 3 clues. Draw the answer. Then check: can a partner draw a DIFFERENT shape that fits your clues? If yes, add a clue.

G8. A long rectangle is cut into 6 squares in a row by 5 lines. How many right angles can you find?

G9. Can a 6-sided shape have 6 right angles inside it? What is the most right angles a 6-sided shape can have inside?

Lesson 9: Measure and Money Masters

Length, mass, volume, money and time • Week 5

START HERE (open with objects, drawing or counting)

S1. A ribbon is 2 m 45 cm long. (1 m is 100 cm.) How many cm is the ribbon?

S2. A bag of rice weighs 3060 g. (1 kg is 1000 g.) Write its mass in kg and g.

S3. A pail holds 4 ℓ 500 ml. A jug holds 1 ℓ 750 ml. (1 ℓ is 1000 ml.) How much more does the pail hold? Hint: write both in ml first.

S4. Ali buys a sandwich for $3.80 and a drink for $1.45. He pays with a $10 note. How much change does he get?

S5. It is 6:20 pm. Write the time in 24-hour time. (After 12 noon, add 12 to the hour.)

S6. A lesson starts at 10:45 am and ends at 12:05 pm. How long is it? Hint: draw a time line and jump to 11:00, then 12:00, then 12:05.

THINK HARDER (the lesson idea in a new setting)

T1. A 2 ℓ bottle of juice is poured into cups. Each cup holds 300 ml. How many cups can be filled? How much juice is left over?

T2. A shop opens at 09:30 and closes at 21:00. How long is it open?

T3. A box with 4 books weighs 1 kg 50 g. The empty box weighs 250 g. All the books weigh the same. How heavy is one book?

T4. Buses leave the interchange every 25 minutes. The first bus leaves at 06:40. The ride to school takes 35 minutes. Ravi must reach school by 08:30. Which is the latest bus he can take?

T5. Ben has only 20¢ and 50¢ coins. He has 9 coins worth $3 altogether. How many of each coin does he have?

T6. A night train leaves at 22:40 and arrives at 01:15 the next morning. How long is the trip?

T7. Siti's home is 1 km 250 m from school. She walks there and back every day. How far does she walk in 2 days?

T8. Ali needs 2 ℓ of milk. A 250 ml carton costs $1.10 and a 1 ℓ carton costs $3.60. What is the cheapest way to buy 2 ℓ?

T9. Mei goes to sleep at 21:45 and wakes up at 06:30. How long does she sleep?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. A 5 m ribbon is cut into two pieces. One piece is 80 cm longer than the other. How long is each piece?

G2. You have a 300 ml cup, a 500 ml cup (no marks on either) and a tap. How can you get exactly 100 ml in a cup?

G3. The clock at the station is 10 minutes fast. The train leaves when this clock shows 13:05. The trip takes 1 h 40 min. What is the real (correct) time when the train arrives? Give it in 24-hour and 12-hour time.

G4. A pail has 4 ℓ 200 ml of water. Ali fills 3 bottles that hold 900 ml each. He pours the rest equally into 2 jugs. How much water is in each jug?

G5. School starts at 08:15. There are 5 lessons of 35 minutes and a 20-minute recess after the 3rd lesson. When does the last lesson end?

G6. Ali and Ben have $24 altogether. After Ali spends $3.60, they have the same amount of money. How much did Ali have at first?

G7. A bottle full of water weighs 1 kg 200 g. After half of the water is poured out, it weighs 750 g. How heavy is the empty bottle?

G8. Mei buys 3 pens and 2 files for $9.20. Each file costs 60¢ more than each pen. How much does one pen cost?

G9. Make up your own problem about money or time that needs at least 2 steps. Write it, solve it and check it. Then change ONE number and solve it again. What changes in the answer, and why? Swap with a partner.

Lesson 10: Puzzle Day

Mixed puzzles: bar graphs and logic • Week 5

START HERE (open with objects, drawing or counting)

S1. Mei is 5th from the front and 4th from the back of a queue. Draw circles for the children in the queue. How many children are in the queue?

S2. Two numbers add up to 50. One number is 12 more than the other. Guess a pair, check it, and improve your guess. What are the two numbers?

S3. Make all the 3-digit numbers you can with the digit cards 1, 4 and 7 (use each card once). Which of them are even?

S4. A bar graph has lines at 0, 10, 20, 30, 40 and 50. A bar stops halfway between 30 and 40. What number does it show?

S5. 5, 9, 13, 17, ... What is the 10th number in this pattern?

S6. Ali has $5. He buys 2 drinks at $1.35 each. How much money does he have left?

THINK HARDER (the lesson idea in a new setting)

T1. Ali has three coins: 10¢, 20¢ and 50¢. He picks one coin, or two coins, or all three. List every different amount of money he can make, in order. How many amounts are there?

T2. Ali, Ben and Siti each have a different pet: a cat, a dog and a fish. Ali's pet has no fur. Ben does not have the dog. Who has which pet?

T3. Triangles are made in a row with sticks: 1 triangle uses 3 sticks, 2 triangles use 5 sticks, 3 triangles use 7 sticks. How many sticks are needed for 12 triangles?

T4. Use the digits 3, 0, 6 and 9 once each. Make the largest even 4-digit number and the smallest odd 4-digit number.

T5. A bar graph shows stamps: Ali 35 and Ben 20. Mei has 15 more than Ben. Siti has as many as Ali and Ben together. How many stamps do the four children have altogether?

T6. Ali, Ben and Siti stand in a line. Ali is not first. Ben is just behind Siti. Who is first?

T7. A snail climbs a wall 12 m high. Each day it climbs up 5 m, and each night it slides down 2 m. On which day does it reach the top?

T8. Two numbers add up to 100. Their difference is 24. What are the numbers?

GO FURTHER (after the topic: extend, justify, make up your own)

G1. Ben is 3rd from the front and 5th from the back of a queue, so there are 7 children. Add the two numbers: what do you get? Does this always work? Mei is 6th from the front in a queue of 15. Where is she from the back? Write a rule that links "from the front", "from the back" and "how many in the queue".

G2. How many squares of any size are there in a 3 by 3 grid of squares?

G3. 6 children meet. Each child shakes hands once with every other child. How many handshakes are there?

G4. One of Ali, Ben and Mei ate the last kueh. Ali says, "Ben ate it." Ben says, "I did not eat it." Mei says, "I did not eat it." Only ONE of them is telling the truth. Who ate the kueh?

G5. Ali, Ben, Mei and Siti read 8, 11, 13 and 15 books (each child a different number). Mei read the most. Siti read more than Ben. Ali read 2 fewer books than Siti. How many books did each child read? Then draw the bar graph.

G6. Squares are built from sticks. Figure 1 is 1 square (4 sticks). Figure 2 is a 2 by 2 block of 4 squares (12 sticks). Figure 3 is a 3 by 3 block of 9 squares (24 sticks). How many sticks does Figure 4 need? Make a table and look at the jumps. How many sticks does Figure 10 need?

G7. Make up your own logic puzzle with 3 people and 3 things (colours, pets or sports). Write 3 or 4 clues. Test it on a partner: does it have exactly ONE answer? Is there a clue you could take away and still have only one answer?

G8. In a queue, Siti is 6th from the front and Ravi is 4th from the back. There are 3 children between them, and Siti is in front of Ravi. How many children are in the queue?

G9. A bar graph shows the paper collected by 4 classes. 3P1 collected 10 kg more than 3P2. 3P3 collected the same as 3P1. 3P4 collected 5 kg less than 3P2. Altogether they collected 135 kg. How much did each class collect?

Answer key (tutor only)

Lesson 1: Number Sleuths

S1. 6049

S2. 800

S3. 1469

S4. 4993

S5. 8000, 8250 (add 250)

S6. 4089, 4098, 4809, 4890

T1. Greatest even: 8520. Smallest odd: 2085 (5 is the only odd card, so it goes last).

T2. 3 numbers: 1001, 1010, 1100.

T3. 8306 (7000 + 1300 + 6)

T4. 5222 (5 + 2 + 2 + 2 = 11)

T5. 1027 (start 1, 0; the last two digits must add to 9 and cannot be 0 or 1, so 2 and 7)

T6. 20 (ten times in the ones place: 5, 15, ..., 95; ten times in the tens place: 50 to 59)

T7. 63, 127 (double and add 1; the jumps are 2, 4, 8, 16, 32, 64)

T8. 10 (3003, 3113, 3223, ..., 3993)

G1. 5137 (137 more than 5000). The closest below 5000 is 3975, which is 1025 less.

G2. No. 14 jumps of 7 is 98, so 1098 is in the pattern. The next number is 1105, so 1100 is skipped.

G3. 252 digits: pages 1–9 use 9, pages 10–99 use 90 × 2 = 180, pages 100–120 use 21 × 3 = 63.

G4. Digit sum 4: 1, 3, 6, 10 (20 numbers altogether). Digit sum 5: the counts by first digit (5, 4, 3, 2, 1) are 1, 3, 6, 10, 15, so 35 numbers altogether. Check: the numbers that start with 4 are 4100, 4010 and 4001 (3 numbers).

G5. 6842 (6 = 2 + 4, 8 = 2 × 4, 6 + 8 + 4 + 2 = 20). It is the only one.

G6. 24 (6 numbers start with each card: for 2, the numbers are 246, 248, 264, 268, 284, 286)

G7. Answers vary. Check that every clue is true for the secret number and that the clues together leave only one number (list the numbers that fit clue 1, then cross out with clue 2, and so on). Example: "I am between 4000 and 5000. My hundreds digit is 1 less than my thousands digit. My ones digit is 3 times my tens digit. My digits add up to 15." The only number is 4326.

G8. Greatest 9810 (9 = 8 + 1 + 0). Smallest 3012 (the smallest sum of three different digits is 0 + 1 + 2 = 3).

G9. 2341. Six numbers start with 1; then 2134 (7th), 2143 (8th), 2314 (9th), 2341 (10th).

Lesson 2: Cryptarithm Crackers

S1. 6455 (3999 + 2456; or add 4000 to get 6456, then take away 1)

S2. 4532 (jumps of 32 + 500 + 4000 = 4532; or 7000 − 2468)

S3. The digit is 6 (4627 + 1365 = 5992)

S4. The digit is 6 (8650 − 3420 = 5230)

S5. 900 (900 + 600 = 1500, and 2400 − 1500 = 900)

S6. 2245 (1275 + 1480 = 2755; 5000 − 2755 = 2245)

T1. A = 0, B = 4, C = 5: 7003 − 4465 = 2538 (check: 2538 + 4465 = 7003)

T2. A = 3, B = 1, C = 3, D = 2: 2371 + 3629 = 6000 (the only way)

T3. Bottom row 1200, 1450, 1900. Middle row 2650, 3350.

T4. 5643 (he added 900 too much: 6543 − 900)

T5. 1602 (Tuesday: 1874 + 650 = 2524; Monday + Tuesday = 4398; 6000 − 4398 = 1602. Check: 4398 + 1602 = 6000)

T6. A = 8, B = 9: 89 + 9 = 98

T7. 6254 and 6257. 10 000 is 1 more than 9999, so the answer is 1 more than 6253. 10 003 is 4 more than 9999, so the answer is 4 more than 6253. (Check: 6254 + 3746 = 10 000.)

T8. A = 9, B = 2, C = 1: 99 + 22 = 121

T9. 9 (35 − 26, or 62 − 53)

G1. A = 9, B = 5, C = 6, D = 3: 8905 − 6437 = 2468 (the only way; check 2468 + 6437 = 8905)

G2. 3087. The only four different digits that add up to 30 are 9, 8, 7, 6, so 9876 − 6789 = 3087.

G3. A = 1, B = 9, C = 8: 11 + 99 + 88 = 198 (the only answer)

G4. A = 7, B = 3, C = 6: 736 − 63 = 673

G5. 4200, with 1400 in the centre. One answer: 1100, 1600, 1500 / 1800, 1400, 1000 / 1300, 1200, 1700.

G6. 981, e.g. 235 + 746 = 981

G7. Answers vary. Check that the sum is true and that the letters are in different columns, so that each column fixes one digit (3 letters in the same column, or a letter in the answer that can be anything, gives many answers). Example: 3517 + 2486 = 6003, rubbed out to A517 + 2B86 = 60C3. The only answer is 3517 + 2486 = 6003.

G8. 247: 5123 − 4876 (the only way). The thousands digits must be next to each other; 5 and 4 in the middle work best.

G9. A = 4, B = 5, C = 9: 459 + 495 = 954

Lesson 3: Times Table Treasure

S1. Both are 48: 8 × 6 and 6 × 8 are the same. Swapping the numbers does not change the answer.

S2. 54 (one more 9: 45 + 9).

S3. 9 boxes (7 × 9 = 63)

S4. 56

S5. 6, 2, 8, 4, 0, 6, 2, 8, 4, 0. Only 6, 2, 8, 4 and 0.

S6. 70 − 7 = 63

T1. 54 and 72 (5 + 4 = 9 and 7 + 2 = 9; 6 + 5 and 8 + 3 make 11). 54 = 9 × 6, 72 = 9 × 8.

T2. 2. 8 × 8 = 64 ends in 4, and 4 × 8 = 32 ends in 2. (8 × 8 × 8 = 512.)

T3. Yes. 6 groups can always be put into 3 pairs of groups, so nothing is left over. (6 × 7 = 3 × 14 = 42.)

T4. 28 and 91. (List 14, 21, 28 ... 98 and add the digits of each: only 28 and 91 make 10.)

T5. (a) 49 is 1 more than 48. (b) 64 is 1 more than 63. (c) 56 is 2 more than 54. The first number of each pair is always more: the closer the two numbers are to each other, the bigger the answer.

T6. 1. Last digits go 9, 1, 9, 1. (9 × 9 × 9 × 9 = 6561.)

T7. 4 or 9 (6 × 4 = 24 and 6 × 9 = 54; no other digit gives a last digit of 4).

T8. 72 every time. For example 5 × 9 + 3 × 9 = 45 + 27; 10 × 9 − 2 × 9 = 90 − 18; 4 × 9 + 4 × 9 = 36 + 36; 8 × 10 − 8 = 80 − 8.

G1. Rows 6 and 7, columns 7 and 8. Start with 49: it can only be 7 × 7.

G2. 4 (18, 48, 78, 108). The last digits 6, 2, 8, 4, 0 repeat every 5, so 8 comes once in every 5.

G3. 11. They are the numbers in the 18 times table: 18, 36, 54 ... 198.

G4. The odd digits 1, 3, 5, 7 and 9 never appear. 8 × any number is even (8 × n = 2 × 4 × n), so its last digit is even, and the last digits cycle through 8, 6, 4, 2, 0. The same is true for the 6 times table (6 × n = 2 × 3 × n, so the last digits cycle 6, 2, 8, 4, 0).

G5. Yes. 6 × 7 = 42 → 2; 2 × 8 = 16 → 6; 6 × 9 = 54 → 4. (The answer is 3024.)

G6. 36 and 72. Numbers in the 6, 9 and 4 times tables together are 36, 72, 108 ..., so the answers are 36 apart. (36 = 6 × 6 = 9 × 4 = 4 × 9; 72 = 6 × 12 = 9 × 8 = 4 × 18. The first number that works is 36, and then it repeats every 36.)

G7. Answers vary. Check that the secret number is in the table, every clue is true, and listing the table leaves exactly one number. Example: "I am a 2-digit number in the 7 times table. I am odd. My digits add up to 10." The 2-digit odd numbers in the 7 table are 21, 35, 49, 63, 77 and 91, and only 91 has digits that add up to 10.

G8. 1. The last digits go 7, 9, 3, 1 and then repeat, so the 4th and the 8th are both 1.

G9. 126. Numbers that work for 7 and 9 are 63, 126, 189 ...; the 3-digit ones below 200 are 126 and 189, and only 1 + 2 + 6 = 9.

Lesson 4: Model Makers

S1. 1165

S2. 213 cm

S3. Siti 373; altogether 885

S4. Ben 92; altogether 115

S5. 387

S6. $12 (the bag costs $15; 4 more boxes of $3)

T1. 48. 4 units = 96, 1 unit = 24, and the difference is 2 units.

T2. 27 (half of the difference).

T3. 250. Take away the extras (50 + 100 = 150): 450 ÷ 3 = 150 in Box B, so A has 200 and C has 250.

T4. Flour 3 kg, rice 6 kg (9 – 3 = 6, 6 ÷ 2 = 3).

T5. 14. The difference, 28 years, never changes. Mother 3 units, Mei 1 unit, so 2 units = 28 and Mei is 14 (mother 42).

T6. $9. The difference, $27, stays the same. Now Ali has 2 units and Ben 1 unit, so 1 unit = $27. Ben now has $27, so Grandma gave each boy $27 – $18 = $9. Check: $54 = 2 × $27.

T7. 60. The difference is 2 units and Ben gave half of it, so 1 unit = 20. Ravi had 20, Ben had 60.

T8. 300 g (1700 – 800 = 900 for 3 equal units, 900 ÷ 3 = 300).

T9. 3 weeks. The difference, $33, stays the same, so Ben must reach $33: $15 + $18. Then Ali has $66.

G1. 10. At first 4 units = 120, so Ben 30 and Ali 90. The total is still 120, but now Ali has 2 parts and Ben 1 part, so 3 equal parts = 120 and each part is 40. Ben now has 40 and Ali 80. Ben went from 30 to 40, so Ali gave 10.

G2. Pen $5, box $10. The second bill is $10 more because it has 2 more pens, so 2 pens cost $10 and 1 pen $5. Then 3 pens cost $15, so the box costs $25 – $15 = $10.

G3. Ravi 35, Ben 50, Ali 100.

G4. Siti 20, Mei 60, Ravi 80. Siti is 1 unit, Mei 3 units, Ravi 3 units and 20 more. Take away the extra 20: 140 for 7 equal units, so 1 unit = 20.

G5. 48 (12 girls and 36 boys at first; 24 and 24 after).

G6. 200 ml. 2 jugs = 6 cups, so 9 cups = 1800 ml.

G7. Answers vary. Check that the model matches the story, the total is the same before and after, and the child can explain what changed. Example: Ali has 50 stickers and Ben has 20. Ali gives Ben some so that they have the same number. (Ali gives 15; the difference is 30 and half of it is 15.) If Ali had 60, he would give 20: the difference grows by 10 and the gift by 5.

G8. 10 years. Now they add up to 10, and each year the total grows by 2 (one year for each child). 30 – 10 = 20, and 20 ÷ 2 = 10. Check: 17 + 13 = 30.

G9. 2 m 5 cm. 475 – 25 = 450 cm for 5 equal units, so 1 unit = 90 cm; third piece = 180 + 25 = 205 cm.

Lesson 5: Remainder Riddles

S1. 8 each, 3 left over (4 × 8 = 32, 35 – 32 = 3)

S2. 8 full bags, 2 sweets left over (7 × 8 = 56)

S3. 7 each, 3 left over (6 × 7 = 42)

S4. 8 cars (30 ÷ 4 = 7 r 2; the 2 children left over need a car too)

S5. 8 pieces, 2 cm left (6 × 8 = 48)

S6. 38 (7 × 5 + 3)

T1. 6 bracelets (53 ÷ 8 = 6 r 5). She needs 3 more beads (56 – 53).

T2. 23 (remainder 3 when ÷ 4: 23 and 27; 23 ÷ 6 = 3 r 5, 27 ÷ 6 = 4 r 3)

T3. 98 (9 × 10 + 8 = 98; 99 ÷ 9 has no remainder)

T4. 12 boatloads (100 ÷ 9 = 11 r 1; the last person needs a boatload too)

T5. 68 oranges (9 × 7 + 5). 68 ÷ 8 = 8 r 4: 8 full bags, 4 left over.

T6. Triangle (50 ÷ 3 = 16 r 2; the 2nd shape in the repeat)

T7. 14, 26 and 38 (each is 2 more than a number in both the 3 and 4 tables: 12, 24, 36)

T8. Bags of 4: 7 bags and 1 left. Bags of 5: 5 bags and 4 left. Bags of 6: 4 bags and 5 left. Bags of 6 leave the most (5), bags of 4 leave the fewest (1).

G1. 62 (60 is in the 3, 4 and 5 tables; 60 + 2 = 62; the next one, 122, has 3 digits)

G2. 72 (the numbers in the 12 times table between 30 and 100 are 36, 48, 60, 72, 84, 96; only 72 leaves remainder 2 when divided by 5: 72 = 14 × 5 + 2)

G3. 7, 14, 21, 28 and 35 (1 r 1, 2 r 2, 3 r 3, 4 r 4, 5 r 5; the remainder cannot be 6 or more)

G4. 73 (35 is in both tables; 35 + 3 = 38 is too small; 70 + 3 = 73)

G5. 3 and 9 (100 ÷ 3 = 33 r 1, 100 ÷ 9 = 11 r 1; 99 is only in the 1, 3 and 9 tables, and dividing by 1 leaves no remainder)

G6. 301 (one more than a number in the 60 times table: 61, 121, 181, 241, 301; only 301 is in the 7 table: 7 × 43)

G7. Answers vary. Check that the number fits both clues and only ONE number does. Example: "Between 20 and 50; ÷ 5 leaves 2; ÷ 6 leaves 4." Only 22 fits (22 = 4 × 5 + 2 = 3 × 6 + 4). With the range 20 to 60, 52 would fit too.

G8. 100 days from today: Wednesday. 100 ÷ 7 = 14 remainder 2, so after 14 weeks it is Monday again, then 2 more days. 100 days ago: Saturday. Go back 14 weeks to a Monday, then 2 more days back: Sunday, then Saturday.

G9. 7 pupils and 38 pens (giving 2 more pens each needs 10 + 4 = 14 pens, and 14 ÷ 2 = 7; 7 × 4 + 10 = 38, 7 × 6 = 42 = 38 + 4)

Lesson 6: Fraction Walls

S1. 5 (1/2 = 5/10)

S2. 1/6 (cutting into fewer slices makes bigger slices)

S3. 7/9 (2/9 + 5/9)

S4. 5/8 (8/8 – 3/8)

S5. 1/12, 5/12, 11/12

S6. 4 (4 small pieces shaded out of 10: 2/5 = 4/10)

T1. 7/12 and 8/12 (1/2 = 6/12 and 3/4 = 9/12; the end fractions are not between). 8/12 = 2/3.

T2. 7/10 (3/5 = 6/10, and 7/10 is bigger than 6/10)

T3. She has read 3/4 (1/4 + 2/4). 1/4 is left.

T4. 7/12, 2/3, 5/6 (7/12, 8/12, 10/12)

T5. 4/6, 6/9 and 8/12 are examples (the top and bottom numbers are both multiplied by 2, 3 or 4, so each piece is cut into more, smaller pieces but the shaded amount does not change; the fraction wall shows it).

T6. 1/3 (5/6 – 3/6 = 2/6 = 1/3)

T7. 3/8 (1/4 = 2/8 and 1/2 = 4/8)

T8. 7/8 (4/8 + 2/8 + 1/8)

T9. No. Both are missing one piece, but 2/3 is missing 1/3 and 3/4 is missing only 1/4, a smaller piece. So 3/4 is bigger (8/12 < 9/12).

G1. 5/9. 3/7 is less than 1/2 (half of 7 is 3 and a half) and 5/9 is more than 1/2 (half of 9 is 4 and a half).

G2. 4/9. 2/5 = 4/10, and ninths are bigger pieces than tenths, so 4/9 > 4/10.

G3. 1/2 = 3/6 and 1/3 = 2/6

G4. 6/12 (= 1/2) and 4/12 (= 1/3). 5/6 = 10/12 and 1/6 = 2/12. Take away the extra 2/12: 8/12 shared between two equal parts is 4/12 each; the bigger one is 4/12 + 2/12 = 6/12.

G5. 8/12 (in 2/3 the bottom is 1 more than the top; 4 times as much: 8/12)

G6. 6/8 (3 + 4 = 7, and 14 is double 7, so double the top and the bottom)

G7. Answers vary. Check that the fraction is equal to the given one and that the clue fits only one fraction. Example: "I am equal to 3/4. My top and bottom numbers add up to 21." Fractions equal to 3/4 are 3/4, 6/8, 9/12, 12/16 ..., with sums 7, 14, 21, 28 ..., so only 9/12 fits. The riddle only works if the number is in the 7 times table (7, 14, 21, 28 ...).

G8. 2 ways: 1/2 + 1/8 + 1/8 + 1/8 + 1/8, and 1/4 + 1/4 + 1/4 + 1/8 + 1/8.

G9. 17/24 (2/3 = 16/24 and 3/4 = 18/24). Another: 25/36 (2/3 = 24/36, 3/4 = 27/36). Many answers are possible: cut the parts smaller and smaller.

Lesson 7: Area Architects

S1. 26 cm (9 + 4 + 9 + 4)

S2. 42 cm² (6 rows of 7)

S3. 8 cm (32 ÷ 4)

S4. 6 squares along one side (6 × 6 = 36), so the distance round is 6 + 6 + 6 + 6 = 24 cm.

S5. 5 rows (8 × 5 = 40). The distance round is 8 + 5 + 8 + 5 = 26 cm.

S6. Area 7 cm². No: the perimeter depends on how the squares are joined (it can be 12, 14 or 16 cm).

T1. 24 by 1 (50 cm), 12 by 2 (28 cm), 8 by 3 (22 cm), 6 by 4 (20 cm). The 6 cm by 4 cm rectangle has the shortest perimeter.

T2. Length + width = 15, so the width is 3 cm. Area 12 × 3 = 36 cm².

T3. Perimeter 20 cm, so the square has sides of 5 cm. Square 25 cm², rectangle 16 cm². The square, by 9 cm².

T4. Each rectangle is 6 cm by 3 cm, so its perimeter is 18 cm.

T5. 5 tiles along and 4 tiles down: 20 tiles.

T6. 6 cm and 3 cm (6 × 3 = 18 and 6 + 3 + 6 + 3 = 18).

T7. Both fences are 28 m. Ali's garden has the bigger area: 49 m² against 45 m².

T8. 10 cm: only the 2 by 3 rectangle. 14 cm: any shape with no 2 by 2 block, e.g. a row of 6. 14 cm is the biggest: 6 squares have 24 sides, they need at least 5 touches to join, and each touch hides 2 sides (24 – 10 = 14).

G1. 11 by 1 (11 cm²), 10 by 2 (20), 9 by 3 (27), 8 by 4 (32), 7 by 5 (35), 6 by 6 (36). The area grows as the sides get closer to each other, and the square has the biggest area. For 40 cm the square is 10 by 10, area 100 cm² (the next best, 11 by 9, is 99 cm²).

G2. The perimeter is 6 sides of a small square, so each side is 6 cm. Area 2 × 36 = 72 cm².

G3. Yes, two answers. A cut parallel to the 4 cm sides is 4 cm long: 20 + 4 + 4 = 28 cm. A cut parallel to the 6 cm sides is 6 cm long: 20 + 6 + 6 = 32 cm.

G4. Small side 5 cm, big side 10 cm. Perimeter 40 cm, area 100 cm² (a small square has area 25 cm²). The sides only double, so the perimeter only doubles (20 to 40). But the area counts all 4 small squares: 4 × 25 = 100 cm².

G5. Corner cut: area 80 − 12 = 68 m², perimeter 36 m (the same as the 10 by 8 rectangle: push the corner out). Notch in the middle of a long side: area 68 m² and perimeter 36 + 3 + 3 = 42 m (take away 4 m of edge and add 4 + 3 + 3). The area is the same in both, but the perimeter is different.

G6. Each small square has sides of 4 cm. The row is 16 cm by 4 cm, so its perimeter is 40 cm. The 8 cm square has a perimeter of 32 cm. The area is 64 cm² for both. The area stays the same but the perimeter goes up when the pieces are spread out.

G7. Answers vary. Check that the clues fit the secret rectangle and that listing the rectangles for the first clue leaves exactly one for the second. Example: "Area 24 cm² and perimeter 20 cm". The rectangles with area 24 cm² are 24 by 1 (perimeter 50), 12 by 2 (28), 8 by 3 (22) and 6 by 4 (20), so only 6 by 4 fits.

G8. Biggest 24 cm (12 sides of 2 cm, e.g. a row of 5). Smallest 20 cm (10 sides: a 2 by 2 square with one more square on a side).

G9. Area 35 − 4 = 31 cm². Perimeter still 24 cm: push the cut corners out and you get the 7 by 5 rectangle back.

Lesson 8: Angle and Line Detectives

S1. 4

S2. The same: a right angle

S3. Smaller

S4. For example H and E (also F, N, Z, U, M).

S5. 4

S6. 3 (2 small ones and 1 made of both)

T1. A rectangle. 2 pairs of parallel lines.

T2. 6 (it has 8 corners; the 2 corners at the bottom of the gap are bigger than a right angle inside the shape).

T3. Smaller than a right angle: 1, 2, 10 and 11 o'clock. A right angle: 3 and 9 o'clock. Bigger: 4, 5, 6, 7 and 8 o'clock (at 6 o'clock the hands make a straight line).

T4. 2 pairs of parallel lines and 4 pairs of perpendicular lines.

T5. (a) For example a shape with a long bottom, a shorter top parallel to it, one upright side and one slanted side. (b) For example a slanted parallelogram or a rhombus with no square corners.

T6. 4: 1 at the top corner, 2 where the middle line meets the upright line, 1 at the bottom corner.

T7. Yes, in both the hands are 4 hour-marks apart the short way round.

T8. 3 pairs: the two upright lines (parallel), the line across with the left upright (perpendicular) and the line across with the right upright (perpendicular). So 1 pair of parallel lines and 2 pairs of perpendicular lines.

G1. 5 lines (4 + 3 + 2 + 1 = 10).

G2. 4 steps: 7. 5 steps: 8. 10 steps: 13. The rule is: number of steps + 3. Each column has one square corner at its top right (that is the number of steps), the tallest column has one more at its top left, and there are 2 square corners at the bottom. The inside corners where one step meets the next are dents (bigger than a right angle), so they do not count.

G3. 36: 4 corners × 1, 8 points on the sides × 2, and 4 crossing points × 4.

G4. The 4th angle is a right angle too (it is a rectangle). Yes: cut one corner off a rectangle with one slanted cut.

G5. No. Perpendicular lines meet at a right angle however the page is turned. On a square grid, one line going 1 right and 1 up at each step and another going 1 left and 1 up at each step are perpendicular (and so are 2 right 1 up with 1 left 2 up). Turning the page so that one line is flat shows the other one standing straight up.

G6. No. A slanted parallelogram (or a rhombus) has 2 pairs of parallel lines but no right angles.

G7. Answers vary. Check that the drawn shape fits every clue and that the clues are about sides, right angles and parallel lines. Example: "I have 6 straight sides. I have 5 right angles inside me. I have one dent." (An L shape.) Other shapes could fit the first two clues, so the third clue matters.

G8. 24. 4 at the corners, and each of the 5 lines makes 2 right angles at the top and 2 at the bottom: 4 + 5 × 4 = 24.

G9. No. The most is 5, for example an L shape. Children try to draw 6 square corners and always find a dent. A proof is not expected.

Lesson 9: Measure and Money Masters

S1. 245 cm

S2. 3 kg 60 g

S3. 2 ℓ 750 ml (4500 ml − 1750 ml = 2750 ml)

S4. $4.75 ($3.80 + $1.45 = $5.25; $10 − $5.25 = $4.75)

S5. 18:20

S6. 1 h 20 min (15 min + 1 h + 5 min)

T1. 6 cups, 200 ml left (300, 600, 900, 1200, 1500, 1800 ml; 2000 − 1800 = 200 ml)

T2. 11 h 30 min (30 min to 10:00, then 11 h to 21:00)

T3. 200 g (1050 g − 250 g = 800 g; 800 ÷ 4 = 200 g)

T4. The 07:55 bus (buses: 06:40, 07:05, 07:30, 07:55, 08:20; 07:55 + 35 min = 08:30; the 08:20 bus arrives at 08:55, too late).

T5. Four 50¢ coins and five 20¢ coins ($2 + $1 = $3)

T6. 2 h 35 min (20 min to 23:00, 1 h to 00:00, 1 h to 01:00, 15 min to 01:15)

T7. 5 km (1250 m × 4 = 5000 m)

T8. Two 1 ℓ cartons: $7.20 (one 1 ℓ and four small: $8.00; eight small: $8.80)

T9. 8 h 45 min (15 min to 22:00, 8 h to 06:00, 30 min to 06:30)

G1. 2 m 10 cm and 2 m 90 cm (500 − 80 = 420; 420 ÷ 2 = 210)

G2. Fill the 300 ml cup and pour it into the 500 ml cup. Fill the 300 ml cup again and pour until the 500 ml cup is full (200 ml goes in). 100 ml is left in the 300 ml cup.

G3. 14:35, which is 2:35 pm (real start 12:55; + 5 min = 13:00, + 1 h = 14:00, + 35 min = 14:35)

G4. 750 ml (3 × 900 = 2700 ml; 4200 − 2700 = 1500 ml; 1500 ÷ 2 = 750 ml)

G5. 11:30 (5 × 35 = 175 min, + 20 = 195 min = 3 h 15 min)

G6. $13.80 ($24 − $3.60 = $20.40; half is $10.20; Ali had $10.20 + $3.60 = $13.80, Ben $10.20)

G7. 300 g (half the water = 1200 − 750 = 450 g; all the water = 900 g; bottle = 1200 − 900 = 300 g)

G8. $1.60 (take away the extra 2 × 60¢ = $1.20: $8.00 for 5 pens; 800¢ ÷ 5 = 160¢. A file is $2.20.)

G9. Answers vary. Check that the problem needs 2 or more steps, the answer is right, and the child can say what changed. Example: "Mei has $20. She buys a $7.85 book and a $5.50 water bottle. Can she also buy 3 stickers at $2.35 each?" She has $20 − $13.35 = $6.65 left, but 3 stickers cost $7.05, so she is 40¢ short. If she had $25 she would have $11.65 left, enough for the stickers, with $4.60 change.

Lesson 10: Puzzle Day

S1. 8 (5 + 4 − 1: Mei is counted in both)

S2. 19 and 31 (19 + 31 = 50 and 31 − 19 = 12; or 50 − 12 = 38 and 38 ÷ 2 = 19)

S3. 147, 174, 417, 471, 714, 741. Even: 174 and 714.

S4. 35

S5. 41 (add 4 each time: 5, 9, 13, 17, 21, 25, 29, 33, 37, 41)

S6. $2.30 ($1.35 + $1.35 = $2.70; $5 − $2.70 = $2.30)

T1. 7 amounts: 10¢, 20¢, 30¢ (10 + 20), 50¢, 60¢ (10 + 50), 70¢ (20 + 50), 80¢ (all three). Listing by the number of coins (one, two, three) makes sure none is missed.

T2. Ali: fish. Ben: cat. Siti: dog.

T3. 25 sticks (add 2 each time: 3 + 11 × 2 = 25)

T4. Largest even: 9630. Smallest odd: 3069.

T5. 145 (Mei 35, Siti 55; 35 + 20 + 35 + 55 = 145)

T6. Siti (the order is Siti, Ben, Ali)

T7. Day 4 (highest point each day: 5, 8, 11, 14 m; it passes 12 m on day 4)

T8. 62 and 38

G1. 3 + 5 = 8, which is 1 more than 7. It always works: the child in question is counted in both numbers, so (from the front) + (from the back) = (children in the queue) + 1. Mei is 15 + 1 − 6 = 10th from the back.

G2. 14 (9 small, 4 of size 2 by 2, 1 of size 3 by 3)

G3. 15 (5 + 4 + 3 + 2 + 1)

G4. Mei. Test each child: if Ali ate it, Ben and Mei are both telling the truth (2 truthful, not 1). If Ben ate it, Ali and Mei are both truthful (2). If Mei ate it, only Ben is truthful (1). So it was Mei.

G5. Mei 15, Siti 13, Ali 11, Ben 8 (the only way). Mei has the most, 15. Ali is 2 fewer than Siti, and the only two numbers that are 2 apart among the rest (8, 11, 13) are 11 and 13, so Siti 13 and Ali 11. Ben is left with 8, and Siti (13) read more than Ben (8).

G6. Figure 4: 40 sticks. The table goes 4, 12, 24, 40, 60, 84, 112, 144, 180, 220. The jumps are 8, 12, 16, 20, 24, 28, 32, 36, 40: they go up by 4 each time. Figure 10 needs 220 sticks. (Rule: $2 \times n \times (n + 1)$.)

G7. Answers vary. Check that the clues fit one arrangement only (try every arrangement of the 3 things). Example: Sam, Lee and Mei wear red, blue and green. Clues: Sam is not wearing red. Lee is not wearing blue. Mei is not wearing green. Sam is not wearing blue. The only answer is Sam green, Lee red, Mei blue. The clue "Mei is not wearing green" could be taken away and there is still only one answer.

G8. 13 (Siti 6th, Ravi 6 + 3 + 1 = 10th from the front, then 3 more behind Ravi)

G9. 3P1 40 kg, 3P2 30 kg, 3P3 40 kg, 3P4 25 kg (take away the extras 10 + 10 − 5 = 15: 120 kg ÷ 4 = 30 kg for 3P2)